

GENODERMATOSIS 2

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Incontinentia pigmenti [blotch-sulzburger syndrome]

- Multisystemic genetic disease
- XLD
- Defect in NEMO gene → increased apoptosis.

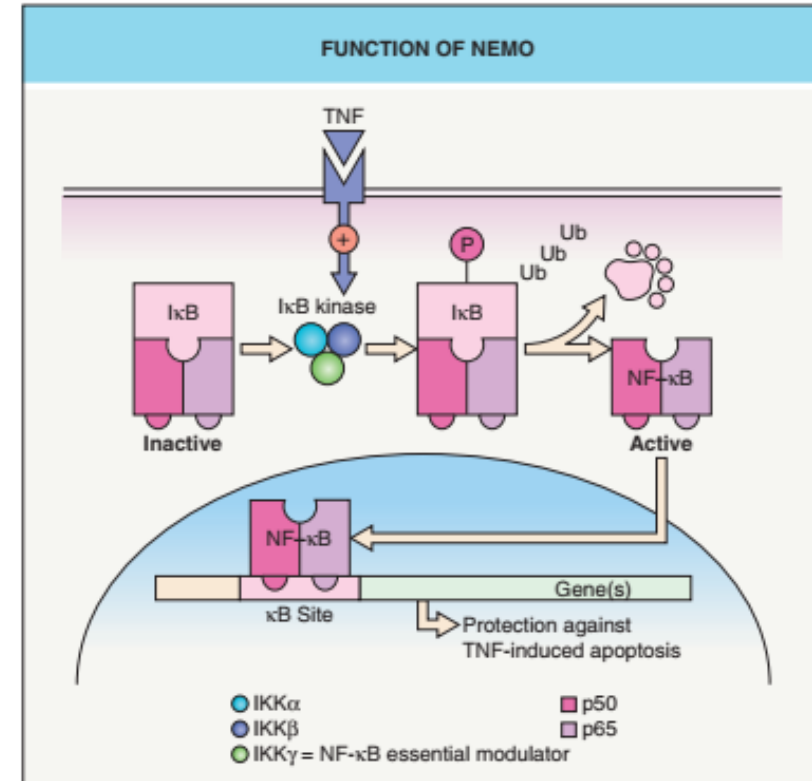
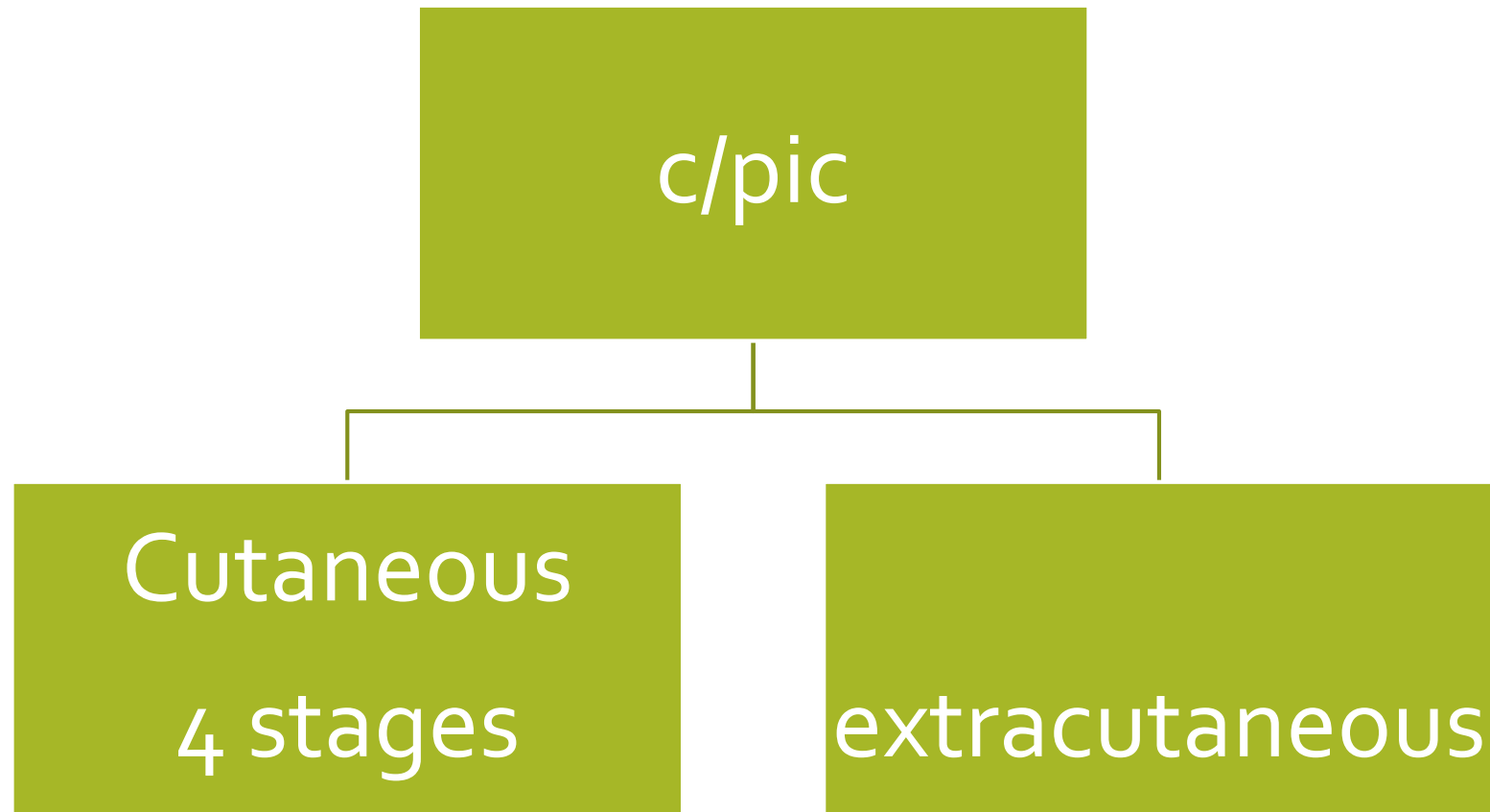


Fig. 62.4 Function of NEMO. In incontinentia pigmenti, lack of NEMO (NF-κB essential modulator; also known as IKBKG [inhibitor of κ light polypeptide gene enhancer in B cells, kinase γ]) results in failure to activate NF-κB, which normally protects against tumor necrosis factor (TNF)-α-induced apoptosis.



Cutaneous 4 stages

- 1) vesicular stage

Erythema

Blister in 1st month in limbs & scalp resolves in days

- 2) Verroucous stage

Linear plaque at extensors (compensatory hyperkeratosis)

- 3) Hyperpigmented stage

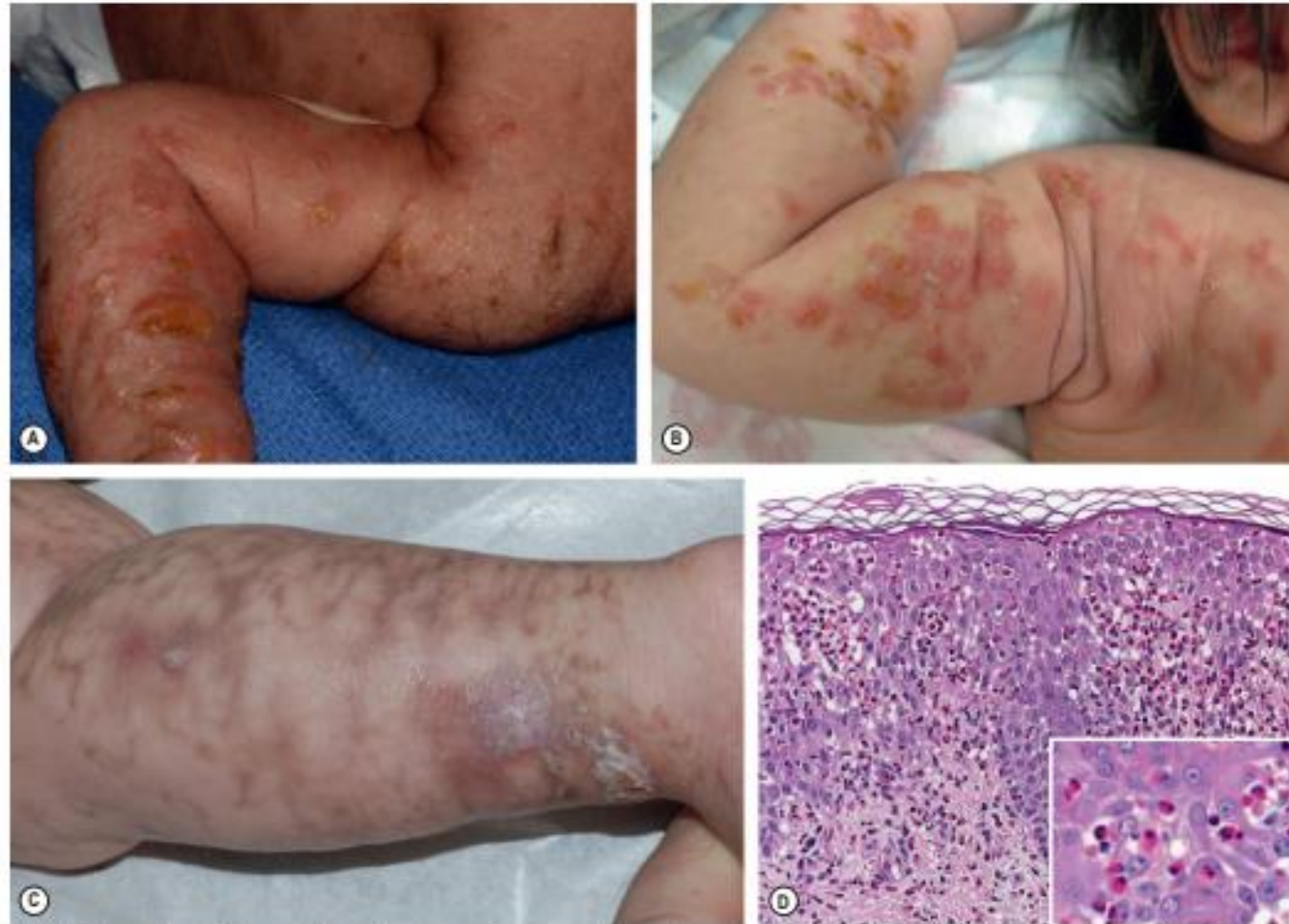
Streaks on trunk & intertrogenous ares (scalloped edges)

- 4) hypopigmented stage :-

bands with decreased hair & sweat gland on calves (only stigmata)



Fig. 62.5 Incontinentia pigmenti. A Stage 1 with streaks of erythema and vesicles in a neonate. B Stages 2 (verrucous) and 3 (hyperpigmented). Note the keratotic lesions on the toes (B). C Residual stage 3 involvement in the axilla of a young girl. The scalloped edges of the hyperpigmented streaks are thought to reflect growth of normal keratinocytes into areas of apoptosis. D Stage 4 (atrophic and hypopigmented). Note absence of hairs within the streaks on the calf. A, Courtesy, Robert A Silverman, MD; B, C, Courtesy, Julie V Schaffer, MD.



eFig. 62.1 Incontinentia pigmenti. A, B Stage 1 presenting as vesicular and crusted lesions. C Stages 2 and 3 with hyperpigmented and hyperkeratotic lesions. D Histologically, numerous eosinophils are present within the dermis and epidermis (inset). Eosinophilic spongiosis can be seen in incontinentia pigmenti as well as disorders such as pemphigus and pemphigoid. A: Courtesy Helen T Shin, MD; B: Courtesy Andrew Zaengle, MD; C: Courtesy Julie H Schaffer, MD; D: Courtesy Lorenzo Cervoni, MD.

Extracutaneous

1. teeth 60-80% missing & pigged & conical
2. nail 10-30 % → ridge, pitting & subangular hyperkeratosis
3. CNS → 20-30 % → seizures & developmental delay.
4. Eye → 20-40 % --. Strabismus & retinal vascular abnormalit
5. Breast, skeletal & pulmonary.

ADDITIONAL MANIFESTATIONS OF INCONTINENTIA PIGMENTI
Surface ectodermal defects
• Missing and conical teeth (40–50% of patients) • Linear absence of hair (swirled at the vertex in 30–40% of patients) and sweat glands • Dystrophic nails (ridging, pitting > subungual keratoses; 10–20% of patients) • Asymmetric breast development or supernumerary nipples
Eye anomalies (20–40% of patients)
• Retinal vascular abnormalities • Strabismus • Cataracts • Microphthalmia • Optic atrophy
Central nervous system abnormalities (~30% of patients)
• Seizures • Motor and intellectual delay • Microcephaly • Hemorrhagic stroke
Skeletal defects (uncommon)
• Skull anomalies • Scoliosis
Other organs (uncommon)
• Pulmonary hypertension

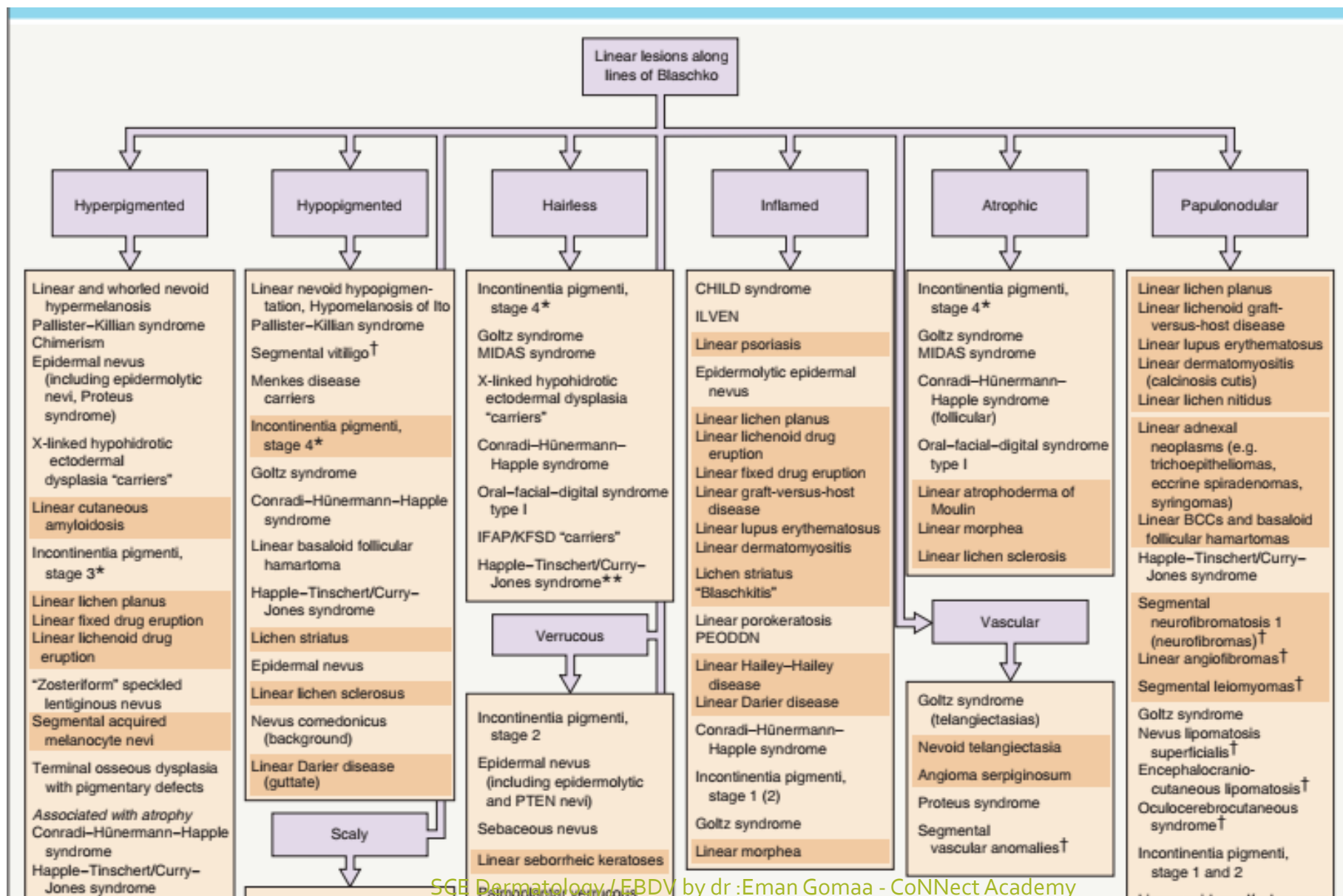
- H.P:- stages

- Inflammatory stage → eosinophilic spongiosis
- Verrucous stage → hyperkeratosis & acanthosis.
- Hyperpigmented stage → pigmentary incontinence

- DD:-

1. Vesiculo-bullous
2. Verrucous → nevus & warts
3. Hyperpigmented → linear whorled nevoid hypermelanosis
4. Atrophic stage → hypomelanosis of ito

- Treatment:-
 - No specific ttt
 - Symptomatic
 - Referral to → neurologist, ophthalmologist & dentist



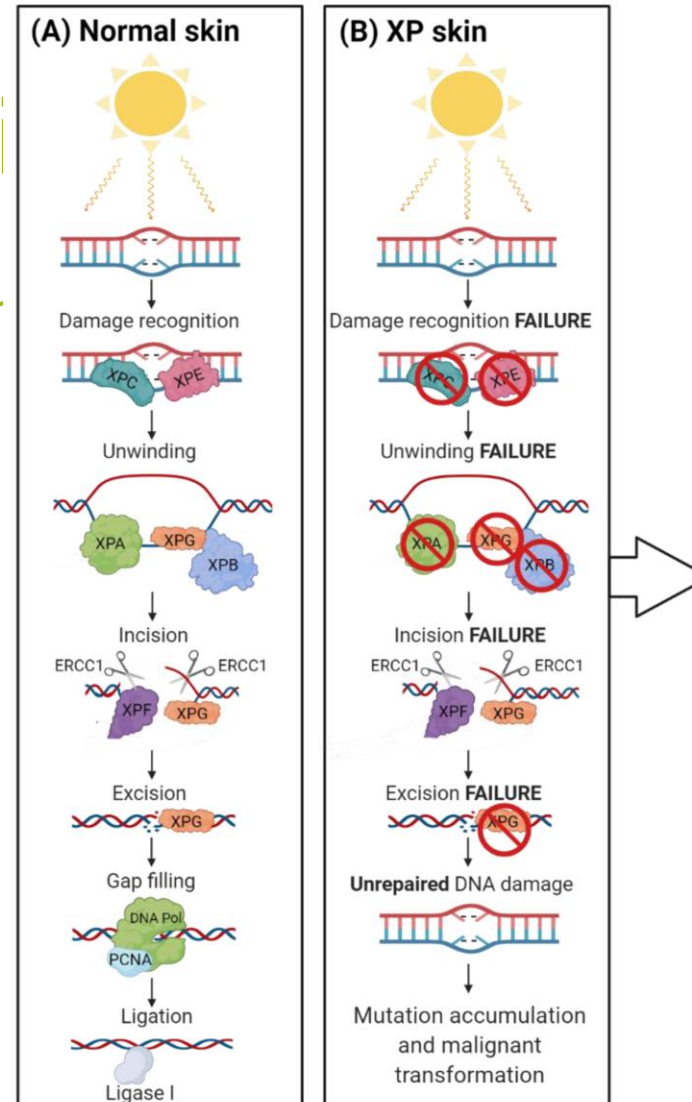
Xeroderma pigmentosa

- Genetic condition characterized by :
- DEFECT IN NUCLEOTIDE EXCISION REPAIR (NER)
- Photosensitivity, pigmentary changes & malignancy
- Rare & equal sex
- Types (7 types XPA → XPG)
- Xerodermoid variant



Aetiopathogenesis

- AR → defect in nucleotide excision repair



c/pic

- 3 findings → Photosensitivity, pigmentary changes & malignancy
- 3 site → skin, mucosa, eye & internal organs



skin

- Photosensitivity → sunburn, erythema
- Pigmentary → solar lentigines & freckles
- Xerosis
- Malignancy → early onset of all types of skin cancer at 8 years
- AK , BCC , SSC , melanoma



- Ocular 40%
 - Dry eye, conjunctival injection & pterygium
 - Madarosis
 - Ectropion

Neurological 20-30%

Not related to UV rays

Oxidative DNA damage → neuronal cell death

Microcephaly, deafness, dysarthria & ataxia

Desancits Cacchione syndrome → most severe form → Mr, deafness, ataxia & quadriparesis

Internal malignancy → 10-20 folds & brain malignancy 50 folds

- Investigations
- Genetic testing
- Neurological invest → deep tendon reflex, audiometry & MRI
- Ttt → restrictive photoprotection
- General measure → stop smoking & eye care & regular screening tests

Cockayne syndrome

- Photosensitivity
- No cancer
- No pigmentary changes
- Bird like facies
- Microcephaly
- Mickey mouse ear
- MR
- Sensorineural deafness
- Defect → NER (TC) → transcription couple



Patient (1)



Patient (3)



Patient (7)

UV sensitive syndrome

- Photosensitivity only
- no cancer
- no neurological
- Ocular → cataract & optic atrophy

Pseudo-xanthoma elastica

- AR

- Defect in ATP binding Cassette [ABCC]

-

-

Physiological Anti-mineralizing factor

-

Its defect results in → organ mineralization by Ca deposition in elastic fibers in Skin, Eye & CVs

C/pic

Skin

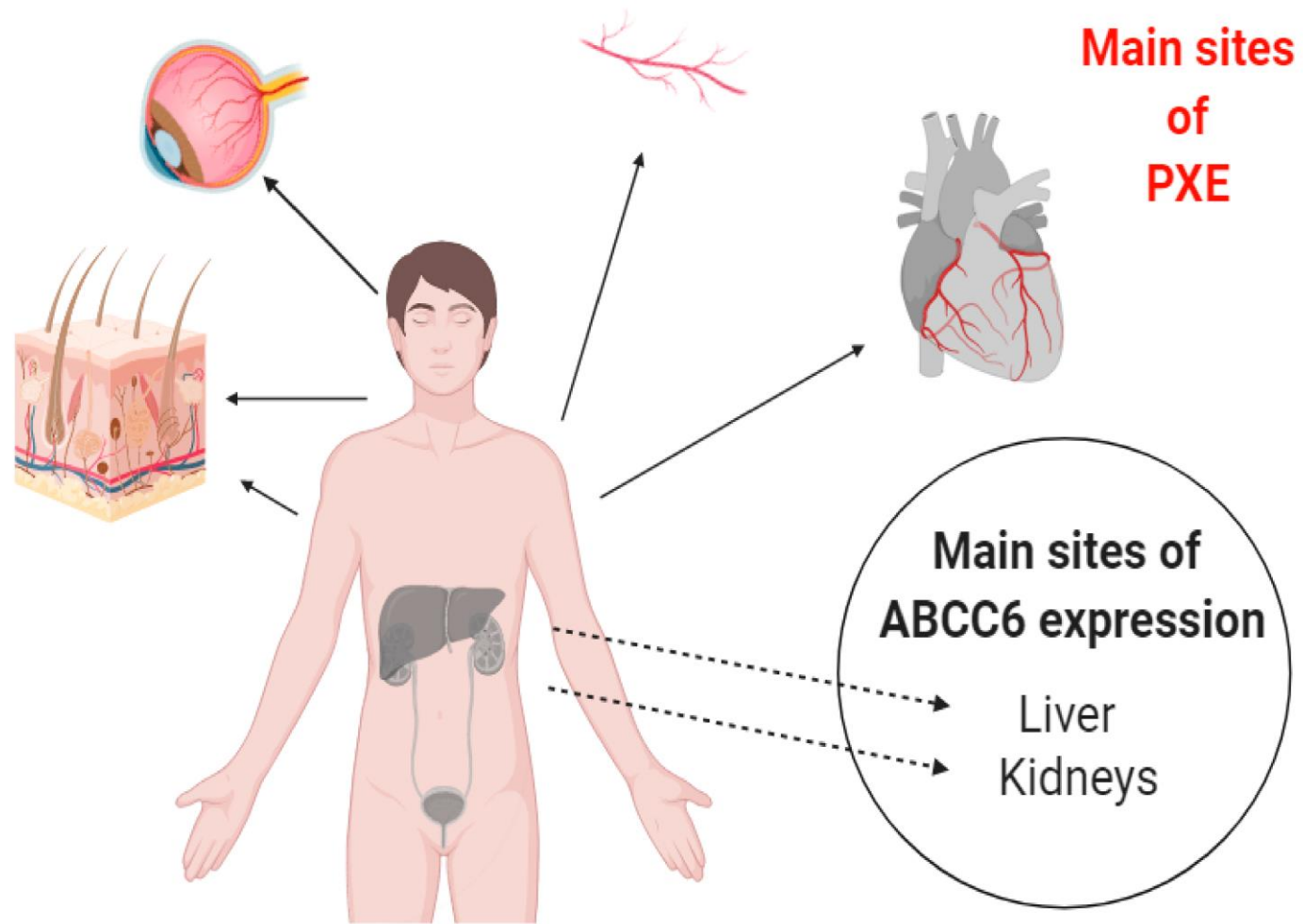
3rd & 4th decade
Yellowish papules →
cobbleston plaque →
plucked chicken skin → in
lateral neck aspects &
flexures

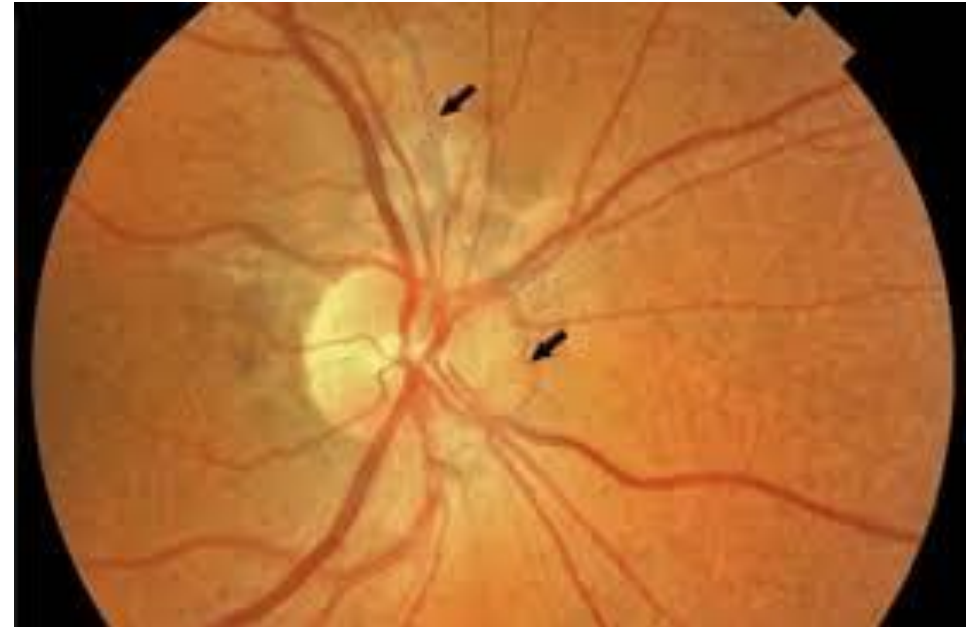
Eye

Bruch membrane →
angeoid streaks → breaks
& calcified bruch
membrane →
neovascularization &
scarring

CVS

Increased morbidity, early
death
Due to calcified media &
intima of BVs
Atheromatous plaque
GIT → bleeding
Extrimities → claudication
Heart → angina & MI





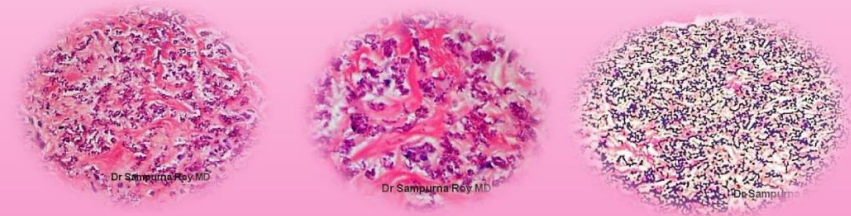


h.p

- Fragmented elastic fiber at mid , deep dermis
- upper dermis → normal
- Verhoeff van giesson stain :elastic fiber
- Von kossa :ca

Pathology of Pseudoxanthoma Elasticum *Dr Sampurna Roy MD*

Microscopic features are characteristic and reveals fragmented and distorted elastic fibers in the reticular and deep dermis. Calcification of fibers can be identified by stains for calcium.
Example: von Kossa stains



histopathology-india.net

DD:

Actinic elastosis

Pxe like papillary dermal elastolysis

White fibrous papules of neck

Acquired pxe

Treatment

- 1- diet :
 - Decrease ca , mg supplement
- 2- skin :
- Eye :
 - Biannual fundoscopy
 - Sun glasses
 - Antioxidant

Defect in tumor suppressor gene

1. Tuberous sclerosis
2. Neurofibromatosis
3. Muri torre
4. Gardner syndrome
5. Couden syndrome
6. Birt hogg dube
7. Reed syndrome

	Tuberous sclerosis	neurofibromatosis
	Defect in tumor suppressor gene → AD , rare Phakomatosis → cutaneous , ocular , CNS	
	TSC ₁ , CHROMOSOME 9 (HAMARTIN) TSC ₂ → chromosome 16 (tuberine)	NF1 → chromosom 17 neurofibromin
	Skin : Facial angiofibroma (nasolabial) Ash leaf (hypopigmented macule (carlist sign Shagreen patch (collagenoma) Guttate leukoderma Peri, subungual fibrom	Skin 1. Neurofibromas :button sign Subcutaneous Plexiform 2- Café au lait macules (calm)earlist sign 3- axillary freckling (crowes sign)

	Tuberous sclerosis	neurofibromatosis
Ocular	Retinal hamartomas Optic glioma Achromatic retinal patch Retinal astrocytoma	Lisch nodules Optic glioma Hypertelorism glaucoma
CNS	Seizures , mr Cortical tubers	Cns tumor , seizure
Systemic	rhabdomyoma	Pheochromocytoma Hypertension

	Tuberous sclerosis	neurofibromatosis
	Angiofibroma :proliferation of fibrous tissue , BV	Neurofibroma :dermal collection of nerve fiber
DD	Vetiligo Nevus anaemicus Nevus depigmentosusu	Calm :mccune –Albright ataxia bloom syndrome



Fig. 61.13 Ash leaf macules in a boy with tuberous sclerosis complex. *Courtesy, [unreadable]*



Fig. 61.14 Guttate leukoderma of tuberous sclerosis complex. Multiple 1–2 mm hypopigmented "confetti" macules. *Courtesy, Jean L. Bolognia, MD.*



Fig. 61.15 Facial angiofibromas and fibrous plaque in tuberous sclerosis complex. **A** Multiple dome-shaped, pink-tan papules on the cheeks and nose of an adolescent. **B** Fibrous plaque on the forehead, which represents the most common location. **A**, Courtesy, Julie V Schaffer, MD; **B**, Courtesy, Ángela Hernández-Martín, MD.



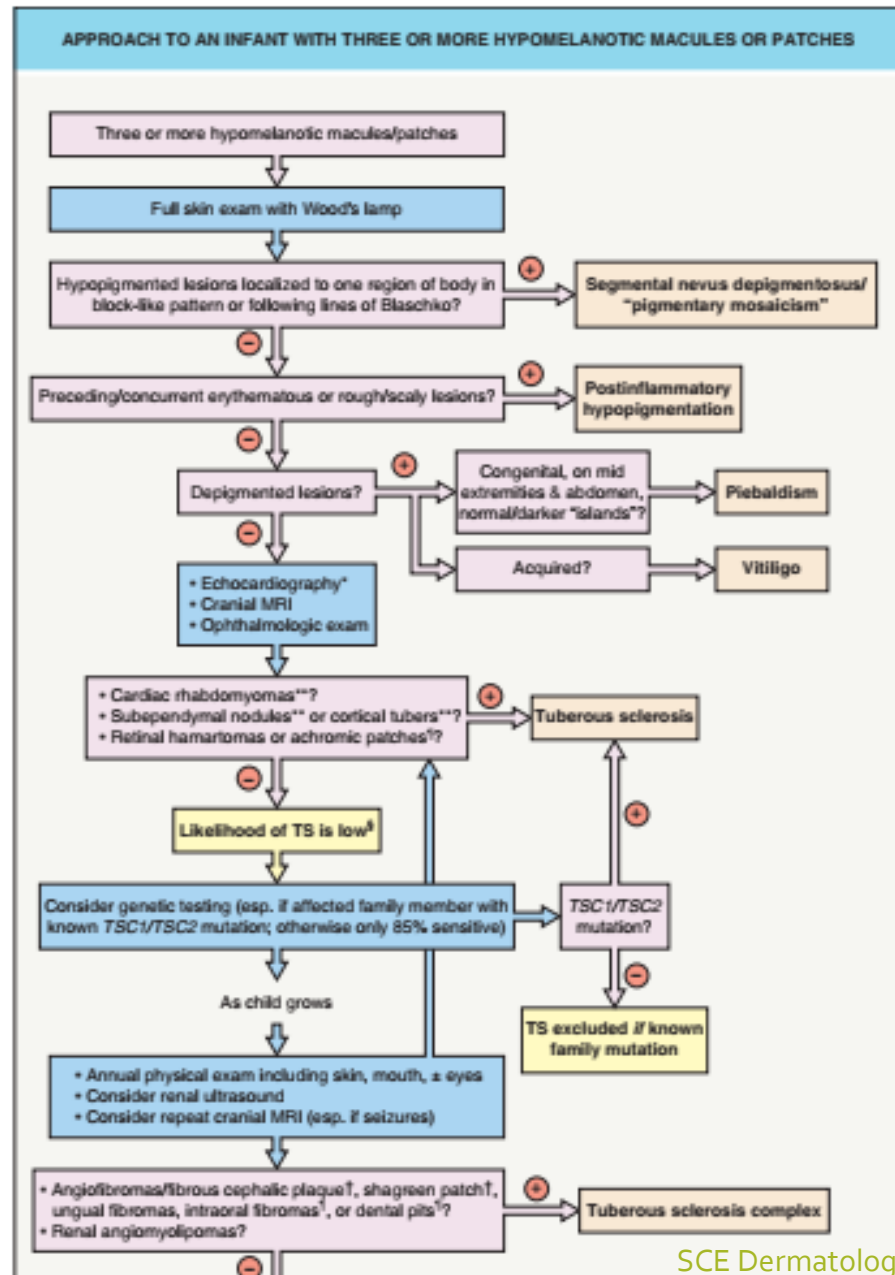
Fig. 61.16 Shagreen patches in tuberous sclerosis. **A,B** Shagreen patches are actually plaques, which can be hyperpigmented, skin-colored (**A**), or hypopigmented (**B**). The surface is said to resemble leather or pigskin. **A&B**



Fig. 61.17 Ungual fibromas of tuberous sclerosis. Multiple fibromas of the toes arising in a peritrichial location. Note the associated longitudinal grooves. Courtesy, Julie V Schaffer, MD.



Fig. 61.18 Retinal hamartomas of tuberous sclerosis. Courtesy, Aaron L. Be



EVALUATION AND MANAGEMENT OF TUBEROUS SCLEROSIS (TSC) PATIENTS

Newly diagnosed

- Neurodevelopmental and behavioral evaluation (especially for children) for features of TSC-associated neuropsychiatric disorder (TAND)
- Brain MRI with and without gadolinium to assess for tubers, migrational defects, subependymal nodules, and SEGA*
- Electroencephalogram (EEG); if abnormal or symptoms of TAND, perform 24-hour video EEG; educate parents of affected infants to recognize infantile spasms
- Ophthalmologic examination including dilated funduscopy
- Electrocardiogram (ECG)
- Echocardiography (in children <3 years of age); consider fetal echocardiography if rhabdomyomas are seen on prenatal ultrasound
- Abdominal MRI to detect angiomyolipomas and renal cysts (favored over CT or ultrasonography)
- Assess blood pressure and determine glomerular filtration rate (GFR)
- High-resolution chest CT and pulmonary function tests (PFTs) to detect lymphangioleiomyomatosis (all women ≥18 years of age; adolescents and men if symptomatic)
- Detailed dental and dermatologic examinations

Established (to monitor for complications)

- Screening for TAND annually; comprehensive formal evaluation for TAND at ages 0–3, 3–6, 6–9, 12–16, and 18–25 years
- Brain MRI every 1–3 years until age 25 to monitor for SEGA; perform MRI more frequently if symptoms or a large/growing SEGA*†
- EEG if known/suspected seizure activity or behavior change/cognitive decline
- Ophthalmologic examination annually if known eye lesions or visual symptoms
- ECG every 3–5 years in asymptomatic patients
- Echocardiography every 1–3 years in asymptomatic pediatric patients with cardiac rhabdomyomas until regression is documented
- Abdominal MRI every 1–3 years to evaluate for progression of angiomyolipomas‡ and renal cysts
- Assess blood pressure and determine GFR at least annually
- High-resolution chest CT every 5–10 years in asymptomatic individuals at risk for lymphangioleiomyomatosis and every 2–3 years together with annual PFTs if known lung cysts
- Dental examination at least every 6 months; panoramic radiographs by age 7 years
- Annual dermatologic examination
- Offer genetic testing/counseling in patients of reproductive age or newly considering having children

Potentially affected family members

- Dermatologic and (if skin findings not diagnostic) ophthalmologic examinations for manifestations of TS
- Consider brain MRI, renal ultrasonography, or (for infants) echocardiography
- Consider genetic testing if a mutation has been identified in the proband

RICCARDI'S CLASSIFICATION OF NEUROFIBROMATOSIS	
Category	Description
NF1	von Recklinghausen disease
NF2	Acoustic schwannoma
NF3*	Mixed (central and peripheral neurofibromas)
NF4*	Variant
NF5*,†	Segmental café-au-lait macules, neurofibromas, or both, affecting one or more dermatomes
NF6*,‡	Café-au-lait macules only
NF7*	Late onset
NF-NOS	Not otherwise specified
*Primarily of historical/conceptual interest. †Now referred to as mosaic (segmental) NF1. ‡Includes Legius (NF1-like) syndrome.	

MAJOR CLINICAL FEATURES OF NEUROFIBROMATOSIS TYPE 1

Cutaneous

- Neurofibromas (80–90%)
- Café-au-lait macules (>90%)
- Axillary and/or inguinal freckling (~80%)
- Plexiform neurofibroma (25%)
- Nevus anemicus (30–50%)
- Juvenile xanthogranuloma (15–35% in first 3 years of life)

Ocular

- Lisch nodules (~90% by 20 years of age)
- Choroidal nodules (>80% of adults)
- Neovascular glaucoma, retinal vasoproliferative tumors

Skeletal

Cranial

- Macrocephaly (20–50%)
- Hypertelorism (25%)
- Sphenoid wing dysplasia (<5%)

Spinal

- Scoliosis (5–10%)
- Spina bifida

Limbs

- Dysplasia of long bone cortex (5%), pseudarthrosis (2%)

Other

- Generalized osteopenia (~50%), osteoporosis (~20%)
- Short stature (~30% <3rd percentile)
- Pectus deformity (30–50%)

Tumors

- Optic glioma (10–15%) (with or without precocious puberty)
- Malignant peripheral nerve sheath tumors (3–15%)
- Pheochromocytoma (~1%)
- Juvenile myelomonocytic leukemia
- CNS tumors other than optic gliomas (~5%)
- Rhabdomyosarcoma, especially of the genitourinary tract
- Duodenal carcinoid
- Somatostatinoma
- Parathyroid adenoma
- Gastrointestinal stromal tumors (GIST)
- Breast cancer (~5-fold increased risk in women <50 years of age)

Neurologic

- Unidentified bright objects (UBO) on MRI (50–75%)
- Learning difficulties (30–50%)
- Seizures (~5%)
- Intellectual impairment (severe in <5%)
- Aqueductal stenosis and hydrocephalus (~2%)

Cardiovascular

- Hypertension (~30%); essential > renal artery stenosis or pheochromocytoma
- Pulmonic stenosis (~1%)
- Renal artery stenosis (~2%)
- Cerebrovascular anomalies (2–5%), including vascular stenoses and aneurysms



Fig. 61.3 Café-au-lait macules & "freckling". Oval-shaped, light- to medium-brown patches with regular borders and uniform pigmentation (A–C). Numerous 1–3 mm lentiginos are most commonly found in the axillae (Crowe's sign) (B) but can also develop in other locations such as the chest (C) and perioral region (D). A–D; Courtesy, Julie V Schaffer, MD.

DISORDERS ASSOCIATED WITH MULTIPLE CAFÉ-AU-LAIT MACULES	
Disorder	Notes
<i>Neurofibromatosis and related disorders</i>	
Neurofibromatosis 1 (NF1)	AD; six or more CALMs in >90% of patients; mutations in <i>NF1</i> gene; see text
Neurofibromatosis 2	AD; six or more CALMs in minority of patients; mutations in <i>NF2</i> gene; see Table 61.8
Mosaic (segmental) NF1 (formerly neurofibromatosis 5)	Neurofibromas alone or CALMs ± “freckling” in a block-like or other mosaic pattern; due to a postzygotic <i>NF1</i> mutation, which may involve the gonads in some patients
Legius (NF1-like) syndrome	AD; six or more CALMs in >80% and intertriginous “freckling” in ~50% of patients; lack neurofibromas, Lisch nodules, optic gliomas, and typical osseous lesions of NF1; macrocephaly and learning disabilities are common; lipomas, hypopigmented macules, and vascular anomalies in minority; loss-of-function mutations in <i>SPRED1</i> gene, the protein product of which inhibits MAPK signaling
Watson syndrome	AD; short stature; pulmonic stenosis; ID; may also have Lisch nodules, axillary/inguinal freckling, and neurofibromas; allelic with NF1; 80/100*
Jaffe–Campanacci syndrome	Disseminated non-ossifying fibromas of long and jaw bones; hypogonadism or cryptorchidism; D; giant cell granulomas (jaws); usually due to <i>NF1</i> mutation
Noonan syndrome	AD; short stature; pterygium colli; characteristic facies; cardiac defects, esp. pulmonic valve stenosis; skeletal and testicular abnormalities; keratosis pilaris atrophicans; lymphedema; melanocytic nevi; activating mutations in <i>PTPN11</i> (allelic with LEOPARD syndrome) > <i>SOS1</i> , <i>RAF1</i> , <i>RIT1</i> > <i>KRAS</i> , <i>NRAS</i> , <i>MAP2K1</i> , <i>CBL</i> , <i>SHOC2</i> (with loose anagen hair), and rarely other genes; patients with an NF1–Noonan overlap phenotype have mutations in <i>NF1</i> gene

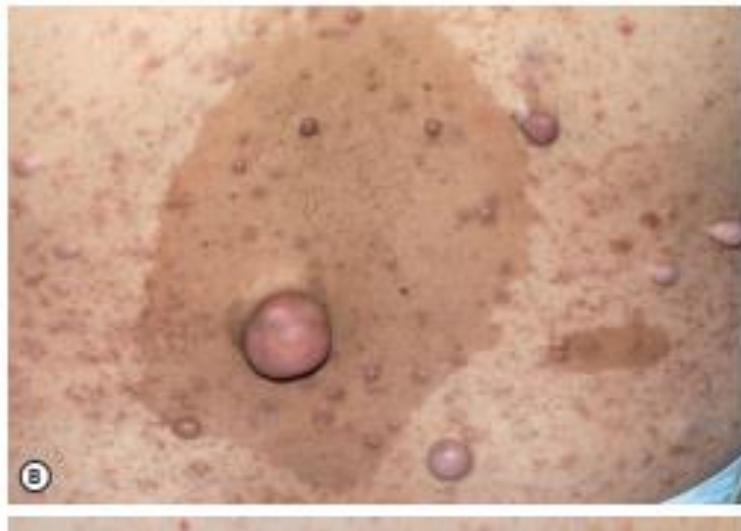


Fig. 61.5 Spectrum of cutaneous neurofibromas. Lesions range from subtle blue-red macules (A) to exophytic nodules with associated hypertrichosis (B). Courtesy, Julie V Schaffer, MD.



Fig. 61.6 Plexiform neurofibromas. Clinical presentations include: hyperpigmented plaques that may be misdiagnosed as a café-au-lait macule (if not palpated) (A) or congenital melanocytic nevus (B); and a poorly circumscribed, sagging pink mass (C). A, Courtesy, Antonio Tanel, MD; B/C, Courtesy, Julie V. Schaffer, MD.



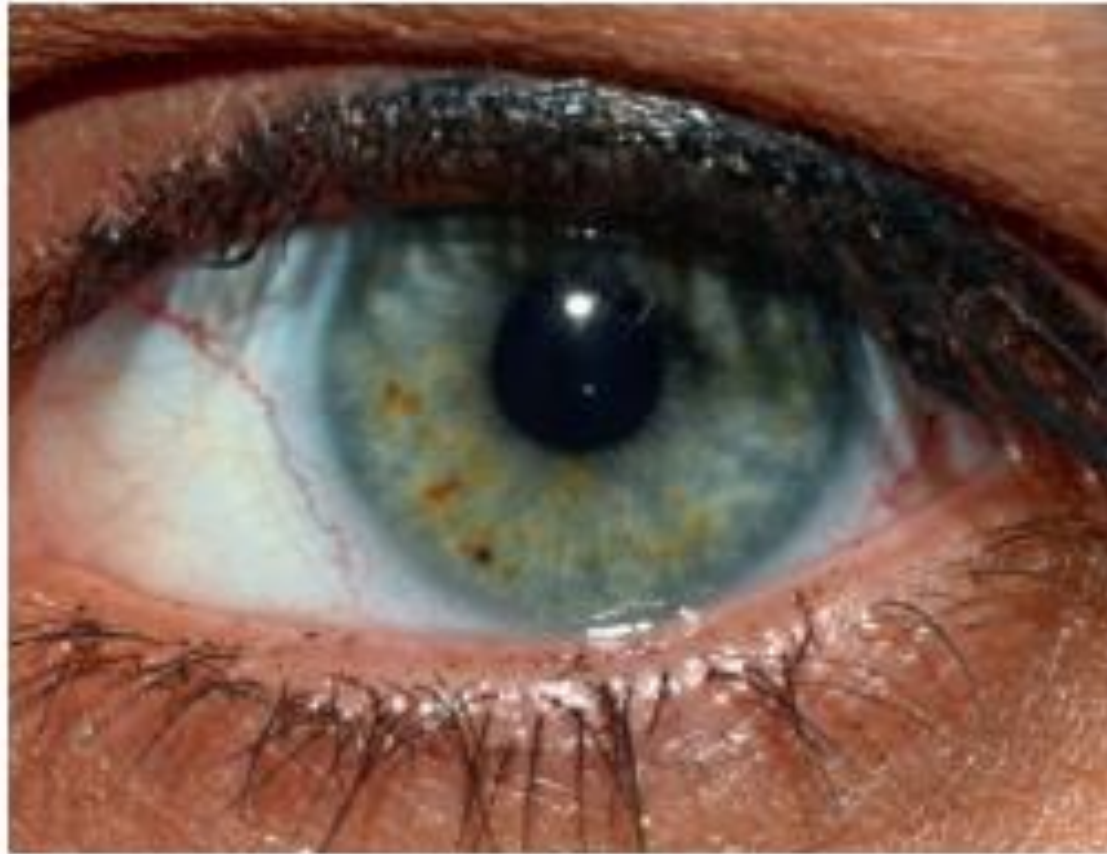
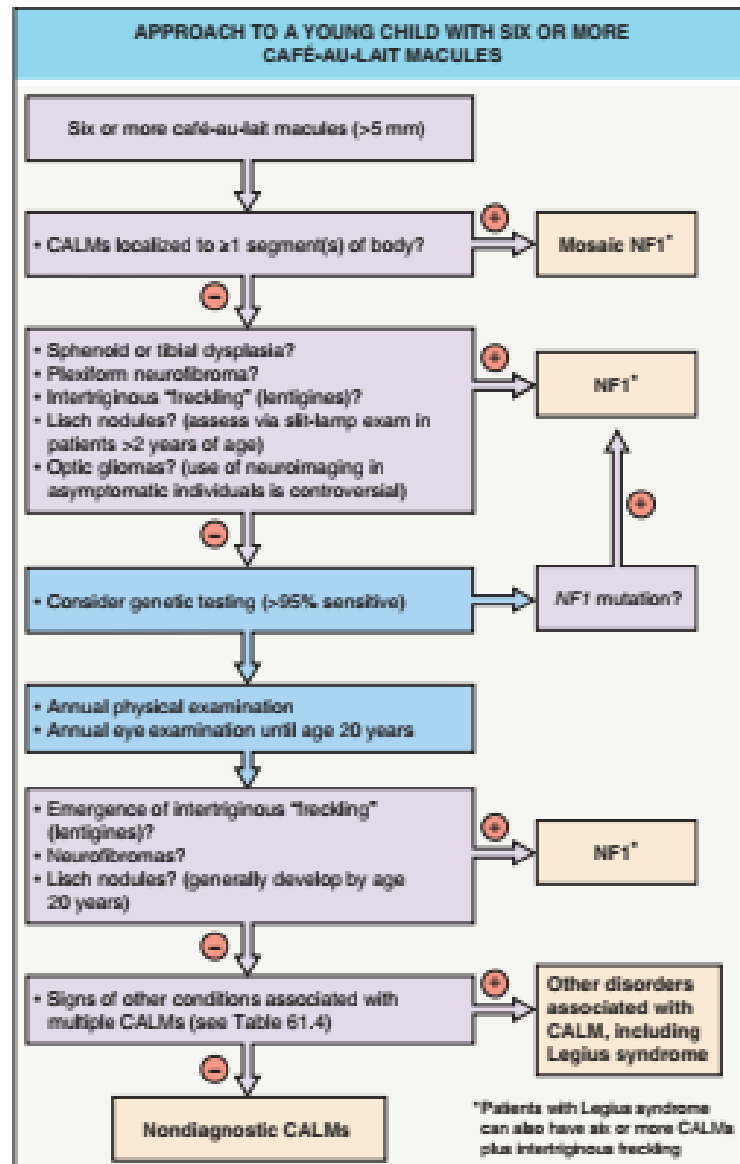


Fig. 61.7 Lisch nodules. Multiple yellow-brown papules on the iris. Although best seen using a slit lamp, Lisch nodules may also be evident on clinical examination in patients with lightly pigmented eyes. *Courtesy, Julie V Schaffer, MD.*

EVALUATION AND MANAGEMENT OF NEUROFIBROMATOSIS 1 (NF1) PATIENTS
At time of diagnosis and annually during childhood and adolescence (unless otherwise indicated) Dermatologic examination (especially if a plexiform neurofibroma [PNF] is present) • Surgical consultation if painful or disfiguring neurofibromas • PET-CT if rapid growth, increased firmness, or persistent pain within an established PNF or the development of a new neurologic deficit* Ophthalmologic examination with visual assessment (prior to age 8 years) • Orbital/brain MRI if signs/symptoms of optic glioma Evaluation for scoliosis and tibial bowing • Referral to orthopedics if scoliosis develops Neurologic examination, developmental/behavioral evaluation, and (in young children) measurement of head circumference • Orbital/brain and/or spinal MRI if neurologic signs/symptoms develop† • Comprehensive developmental assessment prior to starting school and if problems arise Cardiac assessment and measurement of blood pressure • Referral to cardiology if murmur is detected • Renal arteriography and 24-hour urine collection for catecholamines and metanephrines if hypertension develops Assessment for pubertal development • Orbital/brain MRI and referral to endocrinology if precocious puberty develops
Minimum annual evaluation for adults with uncomplicated disease Dermatologic evaluation if PNF is present • PET-CT for indications described above Neurologic examination (especially if a PNF is present) • MRI and/or other studies as indicated if neurologic signs/symptoms develop Measurement of blood pressure • Evaluation as described above if hypertension develops In women, screening for breast cancer with clinical examination + mammography ± MRI (beginning by age 40 years)
Potentially affected family members Dermatologic examination for cutaneous manifestations of NF1 Ophthalmologic examination for Lisch nodules Consider genetic testing if a mutation has been identified in the proband *Immediate assessment is required for any of these findings. †The role of neuroimaging in asymptomatic patients is controversial.



Palmoplantar keratoderma



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graph TD; A[Palmoplantar keratoderma] --> B[hereditary]; A --> C[Acquired]; B --> D[Diffuse]; B --> E[Focal];
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A hierarchical flowchart classifying Palmoplantar keratoderma. The root node is 'Palmoplantar keratoderma', which branches into 'hereditary' and 'Acquired'. The 'hereditary' node further branches into 'Diffuse' and 'Focal'.

hereditary

Acquired

Diffuse

Focal

Diffuse ppk

Isolated

Syndromic



Fig. 58.1 Diffuse palmoplantar keratoderma. Hyperkeratosis of the entire palmar (A) and plantar (B) surface with sharp demarcation. A, Courtesy, Julie V Schaffer, MD; B, Courtesy, Akane Kiro, MD, and David Siegel, MD.

Isolated diffuse ppk

Mal de meleda
AR

NEPPK

Transgradient

Slurp1

Gloves,stocks

Koilonychia

Pseudoainhum

GREITHER
AD

Epidermolytic
Transgradient

K1 gene

Hyperkeratosis

Knee , elbow

Dd ekv

WORNIER

ANASOTHET



Fig. 58.7 Mal de Meleda. Diffuse, yellow hyperkeratosis characterized by maceration, erythema in areas with reduced hyperkeratosis, progression beyond the palmar area (transgrediens), and a well-defined red border.



Fig. 58.5 Diffuse epidermolytic palmoplantar keratoderma (Vörner-Unna-Thost). Diffuse, hyperkeratosis with waxy surface on the palm (A) and soles (B, C), with sharp demarcation. Note the red borders (A, C). C, Courtesy, Allene Koç, MD, and David Siegel, MD.

Diffuse syndromic

Hearing
Wohwinkel
Bart pumphry

Cancer
Huriz
Sex reversal scc

Cardiac
naxos

Ectodermal dysplasia
Clouston
Olmosted

Papillon lefevere

wohwinkl

- AD
- Transgradient
- Loricrin
- K1 : ichthyosis
- GJB2 → CX26 → DEAFNESS
- Honey comb
- Pseudoainhum
- Starfish shape
- Keratosis over knuckles
- Alopecia
- ichthyosis



Fig. 58.8 Mutilating palmoplantar keratoderma in Vohwinkel syndrome. **A** Diffuse honeycombed keratoderma of the sole. **B** Pseudoainhum formation. A, Courtesy, Alfons Kraf, MD, and Dawn Siegel, MD.





Bart pumphry syndrome

- AD
- Knuckles pads
- Leukonychia
- Deafness
- All pts
- GJB2 → cx26
- Unkown,
- Gene



Fig. 58.9 Bart-Pumphrey syndrome. Leukonychia and knuckle pads in a patient with sensorineural deafness. Courtesy, Alfons Krol, MD, and Dawn Siegel, MD.

Huriz

- AD
- SCLERODACTYLY
- NAIL
- Scc
- Complete absence of Langerhans cell
- Palmer > planter



eFig. 58.6 Huriez syndrome. A Mild palmar hyperkeratosis. B Hypoplastic nail changes and sclerodactrophic skin.

Sex reversal scc

- AR
- R-SPONDIN₁
- Nepp
- Sclerodactyly
- Nail dystrophy
- Femal to male sex reversal
- Ambiguous genetalia
- Acral scc
- Early loss of teath

Naxos (cardiac)

- NEPPK
- AR
- PLAKOGLOBINE
- Wolly hair
- Cardiomyopathy
- Non transgradient
- Erythematous border

Clouston syndrome
 AD
 GJB6 → CX30
 Pebbled appearance
 Hypotrichosis
 Nail dystrophy
 eccrine syringofibroadenomatosis



Fig. 58.10 Clouston syndrome. Note the "pebbled" skin on the dorsal aspect of the toes as well as the dystrophic nail plates. Courtesy, D Sasseville, MD, and R Wilkinson, MD.

Olmsted syndrome
 Periorificial keratotic plaque
 AD, AR, XLR
 Flexion deformity
 Constriction digits
 Autoamputation
 SCC
 Melanoma



Papillon Lefevre
 AR, cathepsin
 Loss of teeth
 Scaly psiform lesion
 Pyogenic infection
 Acral melanoma



Focal ppk

Isolated
AD
DG1
DESMOPLAKIN
K6 –K16
K1
c/p :areata
Linear

Syndromic
Howel evan :cancer
Carvajal : cardiac
Ophthalmic :richner –hunhart
Other :pachynychia comgenital

Howel evan

- Neppk
- Pressur area only
- Carcinoma of oesophagus
- 5th decade
- Type A :CANCER , LATE ppk
- TypeB : no ancer early ppk
- DD :Pachynychia congenita



Fig. 58.13 Howel-Evans syndrome. Focal palmoplantar keratoderma in association with carcinoma of the esophagus. Courtesy, Alfons Koç, MD, and Dawn Siegel, MD.

Carvajal syndrom

Demoplakin

Striat eppk

- Wolly hair
- Cardiomyopathy



Fig. 58.14 Carvajal syndrome. A Focal plaque-like hyperkeratosis over pressure points on the soles. B Woolly hair.



Richner –hanhart

- Oculocutaneous tyrosinemia 2
- AR
- Photophobia
- Dendritic keratitis
- Mental retardation

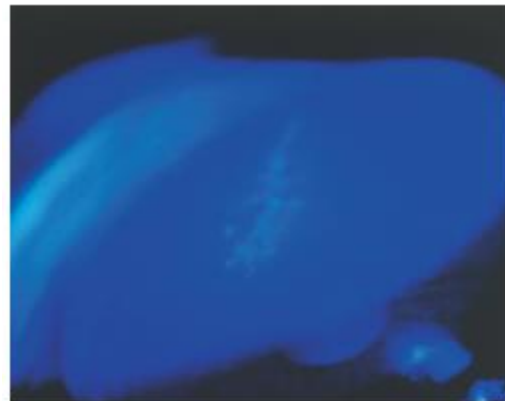


Fig. 58.8 Richner-Hanhart syndrome (tyrosinemia II). Dendritic keratitis detected by slit-lamp examination. Courtesy, J. Bolognia, MD.



Fig. 58.12 Richner-Hanhart syndrome (tyrosinemia II). Focal painful keratoses on the plantar surface in a patient with corneal ulcers and mental retardation. Courtesy, J. Bolognia, MD.

Pachyonychia congenita

- AD
- K6A,K16 –K6B –K17
- NEPPK
- Very painfull sole
- Hyperhidrosis , blistering
- Omega shaped nail
- Natal , perinatal teath
- Oral leukoplaqia
- No malignancy



Palmoplanter keratoderma

PRIMARY PALMOPLANTAR KERATODERMAS (PPKS)	
Disorder	Additional features
Hereditary PPKs	
<i>Diffuse PPK, isolated</i>	
Vörner-Unna-Thost	Epidermolytic
Nagashima, Kimonis	Non-epidermolytic
Bothnia	White/spongy upon water exposure; non-epidermolytic
"Greither"	Transgrediens & progrediens; variable pathology
Mal de Meleda (including Gamborg Nielsen)	Mutilating, frequent superinfections; non-epidermolytic
<i>Diffuse PPK with associated features</i>	
<i>Hearing impairment:</i>	
Vohwinkel syndrome	Mutilating, "honeycomb" PPK; starfish keratoses on knuckles
Bart-Pumphrey syndrome	Knuckle pads, leukonychia
PPK due to connexin 26 defect	"Honeycomb" PPK
Mitochondrial PPK	
<i>High risk of acral SCC*:</i>	
Huriez syndrome	Scleroatrophy
PPK with sex reversal	Female-to-male sex reversal
<i>Hypotrichosis & nail dystrophy:</i>	
Clouston syndrome	Pebbled/grid-like acral papules
Odonto-onycho-dermal dysplasia	Hypodontia, facial telangiectasias/erythema
Schöpf-Schulz-Passarge syndrome	Hidrocystomas, other adnexal neoplasms, hypodontia
Keratoderma-hypotrichosis-leukonychia totalis	Total leukonychia
<i>Mutilating*:</i>	
Olmsted syndrome	Periorificial/intertriginous keratotic plaques
Papillon-Lefèvre syndrome**	Periodontitis, pyogenic infections, psoriasiform plaques
Loricrin keratoderma	Mild generalized ichthyosis, "honeycomb" PPK
KLICK	Keratosis linearis, ichthyosis congenita, sclerosing keratoderma
Vohwinkel syndrome (see above)	
Mal de Meleda (see above)	
<i>Cardiomyopathy:</i>	
Naxos disease***	Woolly hair, arrhythmias in adolescence

<i>Focal PPK, isolated</i>	
Brünauer-Fuhs-Siemens, Wachtlers	Striate on palms, nummular on soles; see Table 58.6 for genetic subtypes
<i>Focal PPK with associated features</i>	
Pachyonychia congenita	Painful soles, dystrophic nails, oral leukokeratosis, steatocystomas/other cysts
Focal palmoplantar and gingival keratosis	Painful soles, gingival leukokeratosis; epidermolytic
Howel-Evans syndrome	Esophageal carcinoma
Richner-Hanhart syndrome (oculocutaneous tyrosinemia)	Dendritic keratitis, corneal ulcers, intellectual disability, painful keratosis
Carvajal syndrome***	Striate on palms; woolly hair, dilated cardiomyopathy; occasionally skin fragility, nail dystrophy, and hypodontia
PPK and woolly hair	Striate on palms; woolly hair, hypotrichosis, leukonychia

Ectodermal dysplasia

Hypohidrotic ED

Hypohidrotic (IMMUNE DEFF-)
XLD
NEMO GENE

HIDROTIC ED
CLOUSTON SYNDROM



Fig. 63.18 Hidrotic ectodermal dysplasia (Clouston syndrome). Note the thickened, shortened nails with distal separation and tiny papules in a regular distribution on the



Fig. 63.16 Male patients with hypohidrotic ectodermal dysplasia. Note the flat nasal bridge, depressed nasal tip, sparse hair (scalp, eyebrows, eyelashes), peg-shaped teeth, full lips and sebaceous hyperplasia. Also note the normal secondary hair in adults. A, Courtesy, Julie V Schaffer, MD; B,D, Courtesy, Mary Williams, MD.

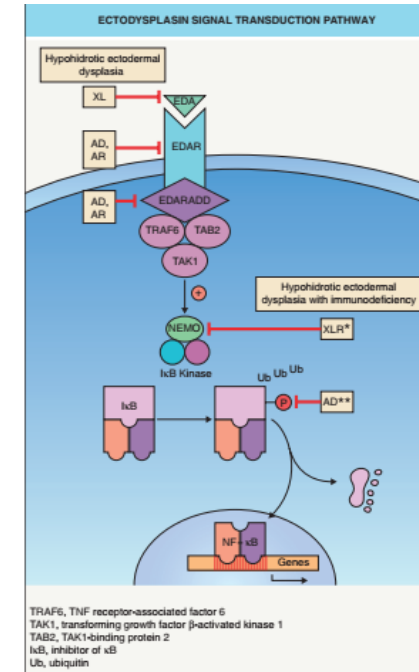


Fig. 63.15 The ectodysplasin signal transduction pathway. *Allelic with incontinentia pigmenti. **A hypermorphic mutation in the gene encoding IκB α that prevents this protein's phosphorylation (and subsequent ubiquitination/degradation) can result in continued inhibition of NF-κB and a phenotype of hypohidrotic ectodermal dysplasia with immunodeficiency. AD, autosomal dominant; AR, autosomal recessive; EDA, ectodysplasin A; EDAR, EDA receptor; EDARADD, EDAR-associated death domain adapter protein; NEMO, NF-κB essential modulator (also known as IKKγ, inhibitor of κ light polypeptide gene enhancer in B cells, kinase γ); XL(R), X-linked (recessive). Additional details of NEMO function are in Fig. 62.4.



Fig. 63.17 Female patients with X-linked hypohidrotic ectodermal dysplasia. Note the peg-shaped teeth, hypodontia, and full lips (A), as well as sparse eyelashes and periorbital hyperpigmentation (B) and sebaceous hyperplasia in a distribution following the lines of Blaschko (C). A, Courtesy, Julie V Schaffer, MD.



